

AIMS System - Detailed Class Design Documentation

28. Design Class: OrderItem

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	productId	Int	null	ID of the product in the order
2	product	Product	null	Reference to the product object
3	quantity	int	0	Quantity of the product ordered
4	unitPrice	float	0.0	Price of the product at time of order
5	subtotal	float	0.0	Total price for this item (unitPrice × quantity)
6	isRushDeliveryEligible	boolean	false	Whether this item is eligible for rush delivery
7	weight	float	0.0	Weight of one unit of this product

Table 2 . Operation design

#	Name	Return type	Description (purpose)
1	OrderItem	void	Constructor for creating a new OrderItem
2	calculateSubtotal	float	Calculates subtotal for this item
3	getTotalWeight	float	Calculates total weight of this item
4	checkEligibilityForRushDelivery	boolean	Checks if eligible for rush delivery
5	toOrderItemDTO	Map<String, Object>	Converts to data transfer object for display
6	getProductId	String	Returns the product ID
7	getQuantity	int	Returns the quantity
8	getUnitPrice	float	Returns the unit price
9	getSubtotal	float	Returns the subtotal
10	getProduct	Product	Returns the product object

Parameter:

- **OrderItem(product: Product, quantity: int)**
 - **product:** The complete Product object
 - **quantity:** Integer representing the quantity of this product in the order
- **OrderItem(productId: String, quantity: int, unitPrice: float, weight: float, isRushDeliveryEligible: boolean)**
 - **productId:** String representing the unique identifier of the product
 - **quantity:** Integer representing the quantity of this product in the order
 - **unitPrice:** Float representing the price at time of order
 - **weight:** Float representing the weight of one unit
 - **isRushDeliveryEligible:** Boolean indicating if the product is eligible for rush delivery
- **calculateSubtotal(): float**
 - **Return:** Float value representing the subtotal (unitPrice × quantity)
- **getTotalWeight(): float**
 - **Return:** Float value representing the total weight (weight × quantity)
- **checkEligibilityForRushDelivery(): boolean**
 - **Return:** Boolean indicating if this item is eligible for rush delivery
- **toOrderItemDTO(): Map<String, Object>**
 - **Return:** Map containing item details for display

Exception:

- **OrderItem(product: Product, quantity: int)**
 - **NullProductException:** Thrown if product is null
 - **InvalidQuantityException:** Thrown if quantity is negative or zero
- **OrderItem(productId: String, quantity: int, unitPrice: float, weight: float, isRushDeliveryEligible: boolean)**
 - **InvalidProductIdException:** Thrown if productId is null or empty
 - **InvalidQuantityException:** Thrown if quantity is negative or zero
 - **InvalidPriceException:** Thrown if unitPrice is negative
 - **InvalidWeightException:** Thrown if weight is negative
- **getTotalWeight(): float**
 - **MissingWeightDataException:** Thrown if weight information is not available

Method:

- **OrderItem(product: Product, quantity: int)**
 - **Purpose:** Creates a new OrderItem from a Product object
 - **Processing:**
 - Validates product and quantity
 - Extracts product details (ID, price, weight)
 - Sets isRushDeliveryEligible based on product.isSupportedPlaceRushOrder
 - Calculates initial subtotal
 - **Side effects:** A new OrderItem is created with initialized attributes
- **OrderItem(productId: String, quantity: int, unitPrice: float, weight: float, isRushDeliveryEligible: boolean)**
 - **Purpose:** Creates a new OrderItem with specific attributes
 - **Processing:**
 - Validates all input parameters
 - Initializes attributes with provided values
 - Calculates initial subtotal
 - **Side effects:** A new OrderItem is created with initialized attributes
- **calculateSubtotal(): float**
 - **Purpose:** Calculates the total price for this order item
 - **Processing:** Multiplies the unitPrice by the quantity
 - **Return:** The calculated subtotal
- **getTotalWeight(): float**
 - **Purpose:** Calculates the total weight of this item for shipping calculations
 - **Processing:** Multiplies the weight by the quantity
 - **Return:** The total weight in kilograms
- **checkEligibilityForRushDelivery(): boolean**
 - **Purpose:** Determines if this item is eligible for rush delivery
 - **Processing:** Returns the value of isRushDeliveryEligible attribute
 - **Return:** True if eligible for rush delivery, false otherwise
- **toOrderItemDTO(): Map<String, Object>**
 - **Purpose:** Creates a data transfer object for display or API responses
 - **Processing:**
 - Collects relevant item data into a map
 - Includes product details, quantity, price information
 - **Return:** Map containing all item details

State

- OrderItem represents a snapshot of a product at the time of order
- It stores both the product ID and optionally a reference to the complete Product object
- The unitPrice reflects the price at the time of order, which may differ from the current product price
- The subtotal is calculated and stored based on quantity and unitPrice
- The isRushDeliveryEligible flag determines whether this item can be delivered with rush service
- The weight attribute stores the product weight for shipping calculations